

NAME: _____

PERIOD: _____

DATE: _____

Two Payloads, One Regression Line

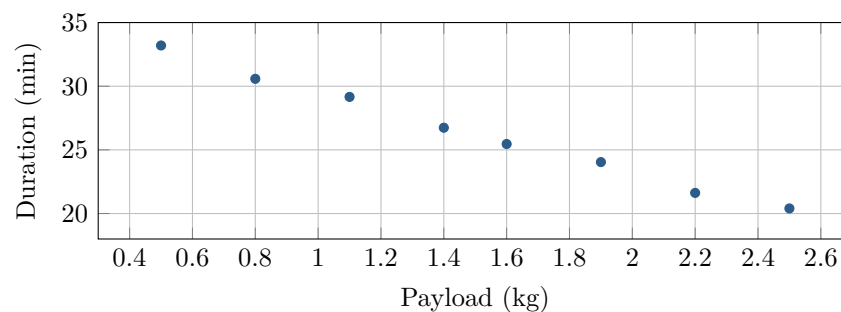
AP Statistics · Unit 5 · Topic 5.3 · B: 2027-02-09 · E: 2027-04-05

[STAR] TODAY'S PROBLEM

Suppose eight drone tests with payloads from 0.5 to 2.5 kg produced the shown data and the model $\hat{y} = 36.0 - 6.4x$, where x is payload and $\hat{y}$ is predicted flight duration in minutes. A proposed mission carries 3.5 kg and requires 12 minutes of flight.

Ravi: Approve it: the model predicts more than the required 12 minutes, and the fitted data are close to a line.

Mina: Before deciding, compare the support for predictions at 1.8 and 3.5 kg and identify evidence the safety decision still needs.



FIRST TAKE — before the video (5 min)

Would you approve now, delay approval, or approve conditionally? Cite one number or plot feature.

[BOOK] USING A REGRESSION MODEL (keep on the page)

A linear model $\hat{y} = a + bx$ predicts response y from explanatory variable x . Its slope is the predicted change in y for a one-unit increase in x ; substitution gives a prediction, not an exact outcome. Interpolation uses an x inside the observed range. Extrapolation uses an x outside that range and becomes less reliable farther from the data because the fitted pattern may not continue. A plausible output does not create evidence beyond the observations used to fit the model.

Finish after the video:

DOK 1 (a) Identify the explanatory and response variables, slope, intercept, and observed payload range.

DOK 2 (b) Interpret the slope. Predict duration at 1.8 and 3.5 kg, classify each prediction, and calculate the modeled buffer above 12 minutes.

DOK 3 (c) Evaluate both approaches using the observed range, the distance from 3.5 kg to that range, the modeled buffer you computed, and the in-range plot. Decide what to do, explain whether the two predictions have equal support, specify evidence needed before approval, and state the consequence of the rejected reasoning.

Frame: At 1.8 kg the model gives _____, supported by _____; at 3.5 kg it gives _____, supported by _____.

Frame: For this mission I would _____ because _____; before relying on the model, collect _____.

EXIT — turn this in

One thing the video changed about my first take: _____